

HOMEWORK 4

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Instructions:

Use this LaTeX file as a template to develop your homework and submit it as a single PDF file on Gradescope. Remember to assign the pages to problems when submitting.

1 Verify PAC Bound

Let h be the VC dimension of function family F . For any $\delta > 0$, with probability at least $1 - \delta$ we have

$$R(\hat{f}_S) - \hat{R}_S(\hat{f}_S) \leq 2\sqrt{\frac{2}{n} \left(h \log n + h \log \frac{2e}{h} + \log \frac{2}{\delta} \right)}$$

where the log is natural, $R(f) = \mathbb{E}1_{[f(x) \neq y]}$ is the risk of f , $S = (x_1, y_1), \dots, (x_n, y_n)$ is a training set of size n , $\hat{R}_S(f) = \frac{1}{n} \sum_{i=1}^n 1_{[f(x_i) \neq y_i]}$ is the empirical risk of f on S , and $\hat{f}_S \in \arg \min_{f \in F} \hat{R}_S(f)$ is an empirical risk minimizer (ERM). We now verify the PAC bound on a simple classification task. Let $p(x) = \text{uniform}([-1, 1])$. Given x , the label is deterministic: $y = \text{sign}(x)$. Recall $\text{sign}(x) = 1$ if $x \geq 0$, and 0 otherwise. This means the true decision boundary is at $x = 0$. Let $f_\theta(x) := \text{sign}(x - \theta)$ which has threshold at θ . Let $F = \{f_\theta : \theta \in [-1, 1]\}$.

1.1 ERM [5 points]

Given S , suppose we find the smallest positive item: $a = \min_{(x_i, y_i) \in S: y_i = 1} x_i$ if one exists, otherwise let $a = 1$. Is $\hat{f}_S := f_a$ an ERM? Justify your answer.

Because the labels are deterministic with $y_i = \text{sign}(x_i)$, every positive example has $x_i \geq 0$ and every negative example has $x_i < 0$.

Let

$$a = \min_{(x_i, y_i) \in S: y_i = 1} x_i$$

if a positive example exists, and $a = 1$ otherwise, and consider $f_a(x) = \text{sign}(x - a)$.

Case 1: at least one positive example. By definition of a , for any positive example $(x_i, 1)$ we have $x_i \geq a$, hence $f_a(x_i) = \text{sign}(x_i - a) = 1 = y_i$. For any negative example $(x_i, 0)$ we have $x_i < 0 \leq a$, so $x_i - a < 0$ and $f_a(x_i) = 0 = y_i$. Thus f_a classifies all training points correctly.

Case 2: no positive examples. Then $y_i = 0$ for all i and we set $a = 1$. Since $x_i \in [-1, 1]$, we have $x_i < 1$ for all i , so $f_a(x_i) = \text{sign}(x_i - 1) = 0 = y_i$ for all i ; again there are no errors.

In both cases

$$\hat{R}_S(f_a) = \frac{1}{n} \sum_{i=1}^n 1_{\{f_a(x_i) \neq y_i\}} = 0,$$

which is the smallest possible empirical risk. Therefore f_a is an empirical risk minimizer and $\hat{f}_S := f_a$ is an ERM.

1.2 True risk [5 points]

Derive $R(f_a)$.

We have $x \sim \text{Unif}([-1, 1])$ and $y = \text{sign}(x)$. Since a is defined as the smallest positive example (or 1), we always have $0 \leq a \leq 1$.

The classifier $f_a(x) = \text{sign}(x - a)$ predicts 1 for $x \geq a$ and 0 for $x < a$.

For $x < 0$ we have $y = 0$ and $x < a$ (because $a \geq 0$), so $f_a(x) = 0 = y$: no error. For $x \geq a$ we have $y = 1$ and $f_a(x) = 1$: no error. Misclassification happens only when $0 \leq x < a$, where $y = 1$ but $f_a(x) = 0$.

Thus the error region is the interval $[0, a)$ of length a inside $[-1, 1]$ of length 2, so

$$R(f_a) = \Pr(f_a(x) \neq y) = \Pr(x \in [0, a)) = \frac{\text{length of } [0, a)}{\text{length of } [-1, 1]} = \frac{a}{2}.$$

So $R(f_a) = \frac{a}{2}$ for $0 \leq a \leq 1$.

1.3 Empirical risk [5 points]

Derive $\hat{R}_S(f_a)$.

From the argument in 1.1, for every training point (x_i, y_i) we have $f_a(x_i) = y_i$, i.e. there are no training errors. Therefore

$$\hat{R}_S(f_a) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{f_a(x_i) \neq y_i\}} = 0.$$

1.4 VC dimension [10 points]

What is the VC dimension of F ? Note F contains threshold classifiers of the type “left negative, right positive.”

The class $F = \{f_\theta : \theta \in [-1, 1]\}$ consists of one-dimensional threshold classifiers of the form $f_\theta(x) = \text{sign}(x - \theta)$ (left negative, right positive).

First, any single point $\{x_1\}$ can be shattered:

- To label x_1 as 0, choose any $\theta > x_1$, so $x_1 < \theta$ and $f_\theta(x_1) = 0$.
- To label x_1 as 1, choose any $\theta \leq x_1$, so $x_1 \geq \theta$ and $f_\theta(x_1) = 1$.

Hence $\text{VCdim}(F) \geq 1$.

Now consider any two points $x_1 < x_2$. Because f_θ is a threshold with $f_\theta(x) = 1$ for all $x \geq \theta$ and 0 for $x < \theta$, it is impossible to have $f_\theta(x_1) = 1$ and $f_\theta(x_2) = 0$ simultaneously: if x_1 is classified as 1, then every larger point x_2 must also be classified as 1. Thus the labeling $(1, 0)$ on $\{x_1, x_2\}$ cannot be realized, so no 2-point set can be shattered and $\text{VCdim}(F) \not\geq 2$.

Therefore $\text{VCdim}(F) = 1$.

1.5 Generalization error [5 points]

Fix $\delta = 0.05$ (95% confidence) and $n = 200$. What is the bound on the true risk of $\hat{f}_S$ given by the PAC bound?

The bound states that, with probability at least $1 - \delta$,

$$R(\hat{f}_S) - \hat{R}_S(\hat{f}_S) \leq 2\sqrt{\frac{2}{n} \left(h \log n + h \log \frac{2e}{h} + \log \frac{2}{\delta} \right)}.$$

For our class F we have VC dimension $h = 1$, and from 1.3, $\hat{R}_S(\hat{f}_S) = \hat{R}_S(f_a) = 0$. With $n = 200$ and $\delta = 0.05$ we obtain

$$R(\hat{f}_S) \leq 2\sqrt{\frac{2}{200} \left(\log 200 + \log(2e) + \log \frac{2}{0.05} \right)}.$$

Using natural logarithms,

$$\log 200 \approx 5.30, \quad \log(2e) \approx 1.69, \quad \log \frac{2}{0.05} = \log 40 \approx 3.69,$$

so the term inside the square root is

$$\frac{2}{200} (5.30 + 1.69 + 3.69) \approx 0.1068.$$

Thus

$$R(\hat{f}_S) \leq 2\sqrt{0.1068} \approx 0.65.$$

So the PAC bound gives a numerical upper bound

$$R(\hat{f}_S) \lesssim 0.65.$$

1.6 Empirical verification [10 points]

Generate 10,000 random training sets from $p(x)$ and the associated labels. Each training set S has $n = 200$ points. On each S you will compute $R(f_a) - \hat{R}_S(f_a)$. Now you have 10,000 numbers: (1) produce a histogram of them, (2) find their 95% quantile of them, and (3) compare the 95% quantile to the bound in the previous question. Discuss your observations.

I simulated the setting 10,000 times. In each repetition I drew a training set S of size $n = 200$ with $x_i \sim \text{Unif}([-1, 1])$ and $y_i = \text{sign}(x_i)$, computed

$$a = \begin{cases} \min\{x_i : y_i = 1\} & \text{if there is at least one positive example,} \\ 1 & \text{otherwise,} \end{cases}$$

formed $f_a(x) = \text{sign}(x - a)$, and then recorded $R(f_a) - \hat{R}_S(f_a)$. Using the results from 1.2 and 1.3, $\hat{R}_S(f_a) = 0$ and $R(f_a) = a/2$, so each value is simply $a/2$.

(1) The histogram of the 10,000 values is highly concentrated near 0 with a long but very light right tail. Most mass lies below about 0.01, and the density decays quickly as the error increases.

(2) The empirical mean of $R(f_a) - \hat{R}_S(f_a)$ over all repetitions is

$$\text{mean} \approx 0.00498,$$

and the empirical 95% quantile is

$$q_{0.95} \approx 0.01507.$$

(3) The PAC bound from 1.5 for $n = 200$, $h = 1$, and $\delta = 0.05$ was

$$R(\hat{f}_S) - \hat{R}_S(\hat{f}_S) \leq 0.65.$$

Comparing, we have

$$\frac{q_{0.95}}{0.65} \approx 0.0232,$$

so the empirical 95% quantile is about 2.3% of the theoretical bound (the bound is roughly 43 times larger). Thus the PAC bound is valid but extremely conservative for this simple threshold-learning problem: in practice the generalization error of the ERM is almost always below 0.02, even though the worst-case, distribution-free bound allows values as large as 0.65.

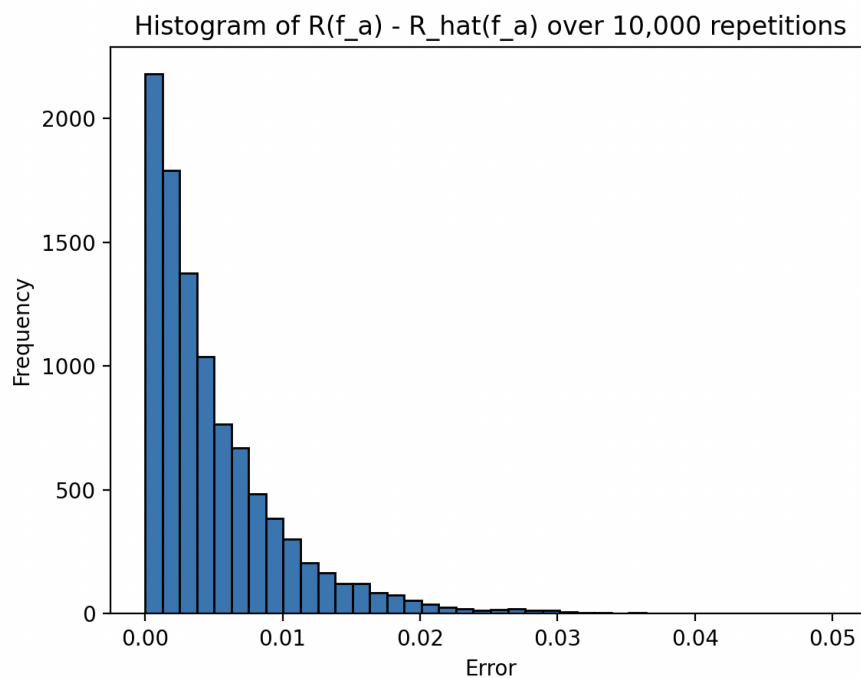


Figure 1: Plot Image

2 VC dimension

2.1 Showing VC dimension [10 points]

Let the input $x \in \mathcal{X} = \mathbb{R}$. Consider $\mathcal{F} = \{f(x) = \text{sign}(ax^2 + bx + c) : a, b, c \in \mathbb{R}\}$, where $\text{sign}(z) = 1$ if $z \geq 0$, and 0 otherwise. Compute the VC-dimension $VC(\mathcal{F})$ of $\mathcal{F}$.

We have $\mathcal{X} = \mathbb{R}$ and

$$\mathcal{F} = \{f(x) = \text{sign}(ax^2 + bx + c) : a, b, c \in \mathbb{R}\}.$$

Each classifier is determined by the quadratic polynomial $g(x) = ax^2 + bx + c$. The sign of $g(x)$ on the real line can change only at real roots of g , and a quadratic has at most two real roots. Hence the real line is partitioned into at most three intervals

$$(-\infty, r_1), (r_1, r_2), (r_2, \infty),$$

on each of which $g(x)$ (and therefore $f(x)$) has a constant sign. If $a > 0$ the sign pattern along the line is $+, -, +$; if $a < 0$ it is $-, +, -$; if there are no real roots the sign is constant.

Lower bound: $V(\mathcal{F}) \geq 3$.

Take any three distinct points $x_1 < x_2 < x_3$. For $a > 0$ the possible label sequences on the ordered points are of the form

$$\underbrace{+, \dots, +}_{n_1}, \underbrace{-, \dots, -}_{n_2}, \underbrace{+, \dots, +}_{n_3}, \quad n_1, n_2, n_3 \geq 0, \quad n_1 + n_2 + n_3 = 3,$$

and for $a < 0$ they are

$$\underbrace{-, \dots, -}_{m_1}, \underbrace{+, \dots, +}_{m_2}, \underbrace{-, \dots, -}_{m_3}, \quad m_1, m_2, m_3 \geq 0, \quad m_1 + m_2 + m_3 = 3.$$

By choosing the roots r_1, r_2 between appropriate points, we can make each block contain the desired number of x_i 's. Checking all sequences of length 3:

$$\begin{array}{ll} +++ : (n_1, n_2, n_3) = (3, 0, 0), & --- : (m_1, m_2, m_3) = (3, 0, 0), \\ ++- : (2, 1, 0), & --+ : (2, 1, 0), \\ +-+ : (1, 1, 1), & -+- : (1, 1, 1), \\ +- - : (1, 2, 0), & -++ : (1, 2, 0), \end{array}$$

so all $2^3 = 8$ possible labelings of $\{x_1, x_2, x_3\}$ are realizable. Therefore any 3 points can be shattered and $V(\mathcal{F}) \geq 3$.

Upper bound: $V(\mathcal{F}) < 4$.

Now consider any four points $x_1 < x_2 < x_3 < x_4$. For any $f \in \mathcal{F}$, as we move from left to right along the real line the sign of $g(x)$ can change at most twice (there are at most two roots). Therefore the label sequence on (x_1, x_2, x_3, x_4) must have at most two sign changes.

However, the labeling pattern $(+, -, +, -)$ has three sign changes $(+ \rightarrow -, - \rightarrow +, + \rightarrow -)$, and similarly $(-, +, -, +)$ has three. These labelings cannot be produced by any quadratic sign classifier. Hence no set of four points can be shattered by $\mathcal{F}$, so $V(\mathcal{F}) \leq 3$.

Combining the lower and upper bounds, we conclude

$$V(\mathcal{F}) = 3.$$

2.2 Shatter Coefficients [10 points]

The main intuition behind VC theory is that, although a collection of classifiers may be infinite, using a finite set of training data to select a good rule effectively reduces the number of different classifiers we need to consider. We can measure the effective size of class $\mathcal{F}$ using *shatter coefficient*. Suppose we have a training set $D_n = \{(x_i, y_i)\}_{i=1}^n$ for a binary classification problem with labels $y_i = \{-1, +1\}$. Each classifier in $\mathcal{F}$ produces a binary label sequence

$$(f(x_1), \dots, f(x_n)) \in \{-1, +1\}^n$$

There are at most 2^n distinct sequences, but often not all sequences can be generated by functions in $\mathcal{F}$. Shatter coefficient $S(\mathcal{F}, n)$ is defined as the maximum number of labeling sequences the class $\mathcal{F}$ induces over n training points in the feature space $\mathcal{X}$. More formally,

$$S(\mathcal{F}, n) = \max_{x_1, \dots, x_n \in \mathcal{X}} \left| \left\{ (f(x_1), \dots, f(x_n)) \in \{-1, +1\}^n, f \in \mathcal{F} \right\} \right|$$

where $|\cdot|$ denotes the number of elements in the set. The *Vapnik-Chervonenkis (VC) dimension* $V(\mathcal{F})$ of a class $\mathcal{F}$ is defined as the largest integer k such that $S(\mathcal{F}, k) = 2^k$.

Suppose there are n points $(x_1, \dots, x_n) \in \mathbb{R}$. Let $\mathcal{F}$ be the collection of 1-d linear classifiers: for each $t \in [0, 1]$, $f_t \in \mathcal{F}$ labels $x \leq t$ as -1 and $x > t$ as $+1$, or vice-versa. What is the shatter coefficient $S(\mathcal{F}, n)$? What is VC-dimension $V(\mathcal{F})$? How can you get it from shatter coefficient?

First sort the n real-valued points so that $x_{(1)} < x_{(2)} < \dots < x_{(n)}$. For any threshold $t \in [0, 1]$, a classifier f_t either labels

$$x \leq t \text{ as } -1, x > t \text{ as } +1 \quad \text{or} \quad x \leq t \text{ as } +1, x > t \text{ as } -1.$$

For a fixed direction “ $x \leq t$ gets -1 ,” the labeling on the ordered points is of the form

$$\underbrace{(-1, \dots, -1)}_{k \text{ times}}, \underbrace{(+1, \dots, +1)}_{n-k \text{ times}}$$

for some $k \in \{0, 1, \dots, n\}$ depending on where t falls among the $x_{(i)}$'s. Thus there are $n+1$ distinct labelings of type “all -1 then all $+1$.”

For the opposite direction “ $x \leq t$ gets $+1$,” the labelings are

$$\underbrace{(+1, \dots, +1)}_{k \text{ times}}, \underbrace{(-1, \dots, -1)}_{n-k \text{ times}}, \quad k = 0, \dots, n,$$

again $n+1$ labelings of type “all $+1$ then all -1 .”

The two families overlap on the two constant labelings (all -1 and all $+1$), so the total number of distinct labelings is

$$S(\mathcal{F}, n) = (n+1) + (n+1) - 2 = 2n.$$

By definition, the VC-dimension $V(\mathcal{F})$ is the largest integer k such that $S(\mathcal{F}, k) = 2^k$. Here $S(\mathcal{F}, k) = 2k$, so we look for k with $2k = 2^k$. This holds for $k = 1$ and $k = 2$, but for $k \geq 3$ we have $2k < 2^k$. Hence the largest such k is $k = 2$, and

$$V(\mathcal{F}) = 2.$$

Equivalently, from $S(\mathcal{F}, k) = 2k$ we see that the class cannot generate all 2^k labelings once $k \geq 3$, so no 3-point set can be shattered, but some 2-point set can, giving VC-dimension 2.

2.3 Monotonicity of VC-dimension [10 points]

Use the shatter coefficients to show the following monotonicity property of VC-dimension: if $\mathcal{F}$ and $\mathcal{F}'$ are hypothesis classes with $\mathcal{F} \subset \mathcal{F}'$, then the VC-dimension of $\mathcal{F}'$ is greater than or equal to that of $\mathcal{F}$.

Assume $\mathcal{F} \subset \mathcal{F}'$. For any fixed choice of n points $x_1, \dots, x_n$, every labeling sequence $(f(x_1), \dots, f(x_n))$ with $f \in \mathcal{F}$ is also realized by some function in $\mathcal{F}'$, because $\mathcal{F}$ is a subset of $\mathcal{F}'$. Therefore, for every n ,

$$S(\mathcal{F}, n) = \max_{x_1, \dots, x_n} |\{(f(x_1), \dots, f(x_n)) : f \in \mathcal{F}\}| \leq \max_{x_1, \dots, x_n} |\{(f'(x_1), \dots, f'(x_n)) : f' \in \mathcal{F}'\}| = S(\mathcal{F}', n).$$

Let $d = V(\mathcal{F})$. By definition, $S(\mathcal{F}, d) = 2^d$, and for all $n > d$ we have $S(\mathcal{F}, n) < 2^n$. From the inequality above,

$$2^d = S(\mathcal{F}, d) \leq S(\mathcal{F}', d) \leq 2^d,$$

so in fact $S(\mathcal{F}', d) = 2^d$. Thus $\mathcal{F}'$ can also realize all 2^d labelings on some d -point set, so $\mathcal{F}'$ shatters d points and

$$V(\mathcal{F}') \geq d = V(\mathcal{F}).$$

Hence the VC-dimension is monotone with respect to set inclusion: if $\mathcal{F} \subset \mathcal{F}'$ then $V(\mathcal{F}') \geq V(\mathcal{F})$.

3 Kernel SVM

Consider the following kernel function defined over $z, z' \in Z$: $k(z, z') = \begin{cases} 1 & \text{if } z = z', \\ 0 & \text{otherwise.} \end{cases}$

3.1 PSD [10 points]

Prove that for any integer $m > 0$, any $z_1, \dots, z_m \in Z$, the $m \times m$ kernel matrix $K = [K_{ij}]$ is positive semi-definite, where $K_{ij} = k(z_i, z_j)$ for $i, j = 1 \dots m$. (Let us assume that for $i \neq j$, we have $z_i \neq z_j$.) Hint: An $m \times m$ matrix K is positive semi-definite if $\forall v \in \mathbb{R}^m, v^\top K v \geq 0$.

For $z_1, \dots, z_m \in Z$ with $z_i \neq z_j$ for $i \neq j$, the kernel matrix $K \in \mathbb{R}^{m \times m}$ has entries

$$K_{ij} = k(z_i, z_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

so K is exactly the $m \times m$ identity matrix I_m .

To show K is positive semi-definite, take any $v \in \mathbb{R}^m$. Then

$$v^\top K v = v^\top I_m v = \sum_{i=1}^m v_i^2 \geq 0.$$

Since this holds for all $v \in \mathbb{R}^m$, the kernel matrix K is positive semi-definite.

3.2 Separability [10 points]

Given a training set $(z_1, y_1), \dots, (z_n, y_n)$ with binary labels, the dual SVM problem with the above kernel k will have parameters $\alpha_1, \dots, \alpha_n, b \in \mathbb{R}$. (Assume that for $i \neq j$, we have $z_i \neq z_j$.) The predictor for input z is

$$f(z) = \sum_{i=1}^n \alpha_i y_i k(z_i, z) + b.$$

Recall the prediction is $\text{sign}(f(z))$. Prove that there exists $\alpha_1, \dots, \alpha_n, b$ s.t. f correctly separates the training set. For a training set $(z_1, y_1), \dots, (z_n, y_n)$ with $y_i \in \{-1, +1\}$ and distinct z_i , the predictor is

$$f(z) = \sum_{i=1}^n \alpha_i y_i k(z_i, z) + b.$$

Because $k(z_i, z_j) = 1$ if $i = j$ and 0 otherwise, for any training point z_j we have

$$f(z_j) = \sum_{i=1}^n \alpha_i y_i k(z_i, z_j) + b = \alpha_j y_j + b.$$

We now choose explicit parameters. Let $\alpha_j = 1$ for all j and $b = 0$. Then, for each training example j ,

$$f(z_j) = 1 \cdot y_j = y_j,$$

and since $y_j \in \{-1, +1\}$,

$$\text{sign}(f(z_j)) = \text{sign}(y_j) = y_j.$$

Thus every training point is classified correctly.

Therefore there exist parameters $(\alpha_1, \dots, \alpha_n, b)$ (for example, $\alpha_i = 1$ for all i and $b = 0$) such that f correctly separates the training set.

3.3 Prediction [10 points]

How does that f predict input z that is not in the training set?

Let z be an input that is *not* in the training set $\{z_1, \dots, z_n\}$. Since all training points are distinct and

$$k(z_i, z) = \begin{cases} 1, & z_i = z, \\ 0, & z_i \neq z, \end{cases}$$

we have $k(z_i, z) = 0$ for all $i = 1, \dots, n$. Therefore

$$f(z) = \sum_{i=1}^n \alpha_i y_i k(z_i, z) + b = \sum_{i=1}^n \alpha_i y_i \cdot 0 + b = b.$$

So for any z not in the training set, the SVM predicts

$$\text{sign}(f(z)) = \text{sign}(b),$$

i.e., all unseen inputs receive the same (constant) label determined solely by the bias term b .

```

hw4.py > ...
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  # Number of datasets
5  T = 10_000
6  n = 200
7
8  errors = np.zeros(T)
9
10 for t in range(T):
11     # 1. Generate uniform samples
12     x = np.random.uniform(-1, 1, size=n)
13     y = (x >= 0).astype(int)
14
15     # Find the smallest positive item
16     pos = x[y == 1]
17     if len(pos) > 0:
18         a = np.min(pos)
19     else:
20         a = 1 # as defined in the homework
21
22     # True risk = a/2 (proven earlier)
23     R_fa = a / 2 # true risk
24     R_hat = 0 # empirical risk is always zero
25
26     errors[t] = R_fa - R_hat
27
28 # ---- Results ----
29
30 # Histogram
31 plt.hist(errors, bins=40, edgecolor='black')
32 plt.title("Histogram of R(f_a) - R_hat(f_a) over 10,000 repetitions")
33 plt.xlabel("Error")
34 plt.ylabel("Frequency")
35 plt.show()
36
37 # 95% quantile
38 q95 = np.quantile(errors, 0.95)
39
40 print(f"Mean error: {errors.mean():.5f}")
41 print(f"95% quantile: {q95:.5f}")
42
43 # Example PAC bound result from section 1.5
44 pac_bound = 0.65
45 print(f"PAC bound from part 1.5: {pac_bound:.5f}")
46 print(f"Ratio (empirical 95% quantile / PAC bound): {q95/pac_bound:.4f}")
47

```

Figure 2: Code Screenshot